
FastPartz

Security & SD-WAN Proposal

Centrally Managed — FlexiWAN + Cloudflare Gateway

Date: February 2026

Version: 2.0

Scope: 4 branches (UAE) + offsite backup

Approach: Cloud-managed SD-WAN with remote administration

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1. Executive Summary

FastPartz operates four branches across the UAE, each running Windows Servers with vendor-supplied accounting software backed by MS SQL Server. Today, these servers sit directly on public IP addresses with RDP and SQL ports exposed to the internet — no firewalls, no antivirus, no backups, and no network segmentation. This configuration represents a critical security risk.

The proposed solution would address every layer of this exposure through a **centrally managed** architecture designed for a business without an on-site IT team. By placing a FlexiWAN SD-WAN appliance at each branch — managed from a single cloud dashboard — encrypting all inter-branch traffic through automated tunnels, filtering DNS traffic via Cloudflare Gateway, and implementing a 3-2-1 backup strategy, the result would be a resilient security posture that can be administered entirely from a remote location.

Key outcomes of the proposed approach:

- **Zero public-facing ports** — full perimeter closure
- **Single cloud dashboard** to manage all 4 branches
- **DNS threat filtering** via Cloudflare Gateway
- **Automated encrypted tunnels** between all branches
- **3-2-1 backups** with immutable offsite copies
- **Full remote administration** — no on-site IT needed

Design principle: This solution prioritizes **manageability and simplicity** over maximum feature depth. A solution the client can operate is more valuable than a complex one that requires expert intervention for every change.

2. Solution Overview

The following table summarizes the proposed changes across each security layer. Every layer is designed to be cloud-managed or self-maintaining.

Layer	Current State	Proposed State
Perimeter	No firewall — public ports open to internet	FlexiWAN firewall at each branch; zero inbound ports
Transport	Unencrypted SQL sync over public internet	AES-256 IPsec tunnels via FlexiWAN SD-WAN (auto-configured)
Segmentation	Flat network — everything reachable	VLANs: servers, workstations, mgmt, guests isolated
DNS Security	No filtering	Cloudflare Gateway: malware, phishing, C2 domains blocked
Endpoint	No AV — Defender disabled	Windows Defender + ESET PROTECT with ransomware rules
Backup	No backups at all	3 copies, 2 media types, 1 offsite, immutable
Access	Direct RDP over internet	WireGuard VPN with key-based authentication
Management	No central visibility	Single cloud dashboard (flexiManage) for all branches

3. Why Cloud-Managed SD-WAN

Traditional firewall appliances (OPNsense, pfSense) are powerful, but each device is managed independently. For four branches with no IT staff on-site, this creates a maintenance burden that grows with every configuration change.

Concern	Traditional Firewall	FlexiWAN SD-WAN
Management	4 separate web GUIs — one per branch	Single cloud dashboard for all branches
VPN tunnels	Manual IPsec on each end: 12 configurations	Select devices, click “Create Tunnels”, done
Updates	SSH into each device individually	Push from cloud to all devices at once
Troubleshooting	Need SSH access + networking expertise	Cloud dashboard shows device and tunnel status
IP changes	Manual DynDNS + tunnel re-keying	Handled automatically by cloud controller
Remote admin	Need VPN just to reach each firewall GUI	Web dashboard accessible from anywhere

FlexiWAN runs on the same Intel N100 mini-PC hardware as a traditional firewall. The difference is operational: all configuration, monitoring, and troubleshooting happens through a central cloud console rather than per-device.

Trade-off: FlexiWAN's built-in firewall is simpler than OPNsense and does not include IDS/IPS (intrusion detection). This gap is addressed by Cloudflare Gateway for DNS-layer threat blocking and ESET PROTECT for endpoint defense. The combination covers the critical threat vectors for this environment.

4. Recommended Software & Cost

All core infrastructure uses open-source software. The primary recurring costs are FlexiWAN cloud management and enterprise antivirus.

Software	Purpose	Key Features
FlexiWAN	SD-WAN + Firewall	Cloud-managed, automated IPsec tunnels, app identification, VLAN support, DHCP, firewall rules
Cloudflare Gateway	DNS Security	Blocks malware/phishing/C2 domains, DNSSEC, content categories (free for 50 users)
WireGuard	Remote Access VPN	Fast, modern, native clients on all platforms, key-based authentication
Windows Defender	Endpoint AV	Real-time protection, Tamper Protection, ASR rules, cloud-delivered protection
ESET PROTECT	Enterprise AV	Central cloud console, ransomware shield, behavioral detection
Veeam Backup CE	Server Backup	Full/incremental, MS SQL VSS, application-aware, AES-256 encryption (free ≤10 workloads)
Restic	Cloud Backup	Backblaze B2 integration, encryption, deduplication, retention policies
Uptime Kuma	Monitoring	Ping/TCP/HTTP checks, web dashboard, Telegram + email notifications

Hardware per Branch

Each branch would require two devices:

- **FlexiWAN Edge (N100 Mini-PC)** — Intel N100, 4× 2.5GbE I226-V NICs, 8 GB DDR5, 128 GB NVMe, fanless. Runs Ubuntu 20.04 + FlexiWAN. (\$150–200)
- **Managed Switch** — TP-Link TL-SG2428P or similar with 802.1Q VLAN support. For smaller branches, an 8-port model is sufficient. (\$200–300)

Optional: Backup NAS at the master branch (\$400–500), offsite NAS (\$400–500).

Cost Summary

Scenario	Hardware	Year 1 Recurring	Year 1 Total
Core — 4× N100 + switches	\$1,520–2,200	\$2,080	\$3,600–4,280
Core + self-hosted mgmt	\$1,520–2,200	\$280	\$1,800–2,480

Core + Backup NAS	\$1,920–2,700	\$2,080	\$4,000–4,780
Full (Core + NAS + offsite)	\$2,320–3,200	\$2,080	\$4,400–5,280

Recurring breakdown: FlexiWAN cloud: \$160/month (\$1,920/year) + ESET: \$160/year = \$2,080/year.

Self-hosted option: Host flexiManage on a \$10/month VPS, reducing total recurring to ~\$280/year.

5. Network Architecture

Each branch follows the same topology: broadband router hands off to a FlexiWAN appliance, which serves as the gateway for all traffic. A managed switch provides VLAN segmentation.

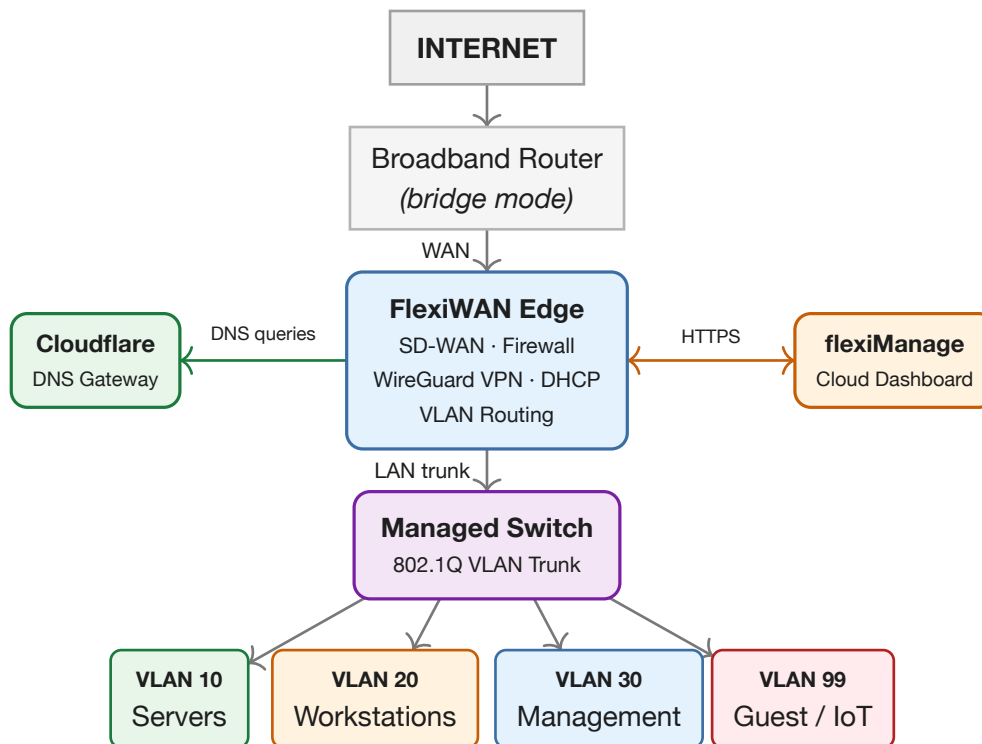


Figure 1: Proposed per-branch topology with cloud management and DNS security

Traffic Type	Path	Protection
Server → Server (inter-branch)	FlexiWAN → IPsec SD-WAN tunnel → remote branch	AES-256 encrypted (auto-configured)
Workstation → Internet	FlexiWAN → ISP	DNS filtered by Cloudflare Gateway
Workstation → Server	FlexiWAN inter-VLAN routing	Firewall rules + Windows Firewall
Remote user → Server	WireGuard VPN → FlexiWAN → Server	Key-authenticated + encrypted
Any device → DNS	FlexiWAN → Cloudflare Gateway	Malware/phishing/C2 blocked

6. Network Segmentation

Why Segment the Network?

In a flat network, a single compromised device — whether a workstation hit by phishing or a printer with a default password — can reach every other device on the network. This is how ransomware spreads: once inside, it moves laterally to find servers, databases, and backups.

Network segmentation through VLANs creates isolated zones with strict rules between them. Even if an attacker compromises a workstation, they would be unable to reach servers in a different VLAN without passing through FlexiWAN, which enforces explicit access rules. This dramatically limits the blast radius of any incident.

IP Addressing Scheme

Pattern: 10.<branch>.<vlan>.0/24

Branch	VLAN 10 (Servers)	VLAN 20 (Workstations)	VLAN 30 (Mgmt)	VLAN 99 (Guest)
Branch 1	10.1.10.0/24	10.1.20.0/24	10.1.30.0/24	10.1.99.0/24
Branch 2	10.2.10.0/24	10.2.20.0/24	10.2.30.0/24	10.2.99.0/24
Branch 3	10.3.10.0/24	10.3.20.0/24	10.3.30.0/24	10.3.99.0/24
Branch 4	10.4.10.0/24	10.4.20.0/24	10.4.30.0/24	10.4.99.0/24

VLAN	Name	Devices	Access Rules
10	Servers	Windows Servers, NAS, SQL	Reachable from workstations on SQL + file share ports only. Inter-branch via SD-WAN.
20	Workstations	User PCs, laptops	Can reach servers for SQL and file shares. Full inspected internet access.
30	Management	FlexiWAN, switch mgmt	IT administrators only. All other VLANs blocked.
99	Guest / IoT	WiFi guests, printers, cameras	Fully isolated. Internet-only (HTTP/HTTPS). Cannot reach internal VLANs.

7. Security Layers

The proposed architecture combines network-level and endpoint-level defenses. Each layer compensates for the limitations of the others.

Defense Layer	What It Does	Priority
Firewall	FlexiWAN default-deny inbound. Closes all public ports — RDP, SQL, everything.	Critical
DNS Filtering	Cloudflare Gateway blocks malware, phishing, C2, and DGA domains before they resolve.	Critical
VLAN Segmentation	Isolates servers, workstations, management, and guests behind access rules.	Critical
Encrypted Transport	FlexiWAN IPsec tunnels with AES-256. Auto-configured, auto-recovering.	Critical
Endpoint AV	Windows Defender + ASR rules + ESET PROTECT enterprise antivirus.	High
Server Hardening	Disable SMBv1, NetBIOS, LLMNR, PowerShell v2. Enable Windows Firewall, audit logging.	High
Application Control	AppLocker blocks executables from user profile, temp, and AppData folders.	Medium
Backup Immutability	3-2-1 backup with Object Lock. Ransomware cannot encrypt backup copies.	Critical
Monitoring	Uptime Kuma dashboard + Telegram alerts for outages and failures.	High

8. Encrypted Connectivity

The accounting software currently syncs SQL data over the open internet. The proposed solution would route all inter-branch traffic through encrypted SD-WAN tunnels that configure themselves automatically.

Feature	Specification
Protocol	IPsec over VXLAN (FlexiWAN native)
Encryption	AES-256 (keys auto-generated by flexiManage)
Tunnel Setup	Select devices in dashboard → “Create Tunnels” → full mesh auto-configures
Route Advertisement	LAN subnets automatically shared across tunnels
NAT Traversal	Automatic — works behind ISP routers
Health Monitoring	Real-time tunnel status in flexiManage dashboard
Recovery	Automatic reconnection on failure — no manual intervention

Full-Mesh Topology

All four branches would be connected via six encrypted tunnels, providing direct branch-to-branch communication.

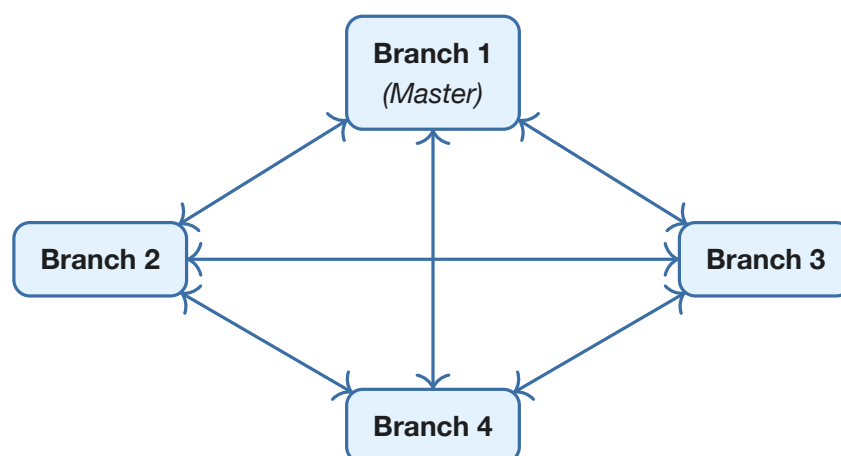


Figure 2: Full-mesh SD-WAN — 6 encrypted tunnels, auto-configured from a single dashboard

SD-WAN beyond basic VPN: FlexiWAN provides application identification, link quality monitoring, and path selection — enabling SQL sync traffic to be prioritized over general browsing, and automatically re-routed if link quality degrades.

9. Backup Strategy (3-2-1)

Three copies of data, on two different media types, with one copy offsite. Backblaze B2 is recommended for the cloud tier.

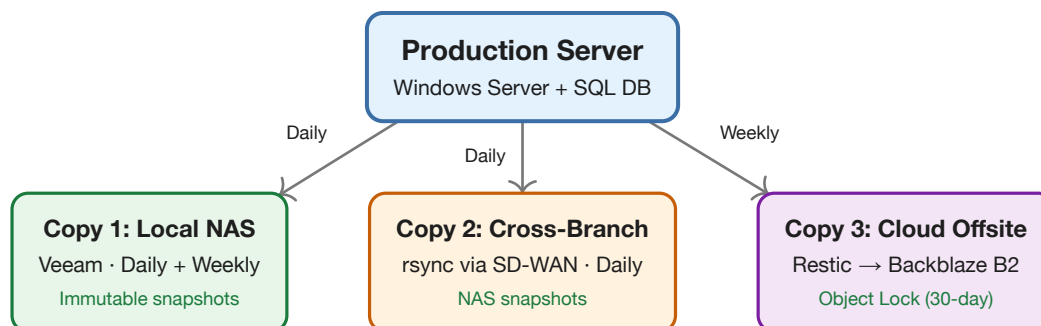


Figure 3: 3-2-1 backup architecture with immutability at every tier

Copy	Location	Method	Schedule	Immutability
1 — Local	Synology NAS (master branch)	Veeam CE (SQL-aware, free)	Daily incremental + weekly full	NAS snapshots (7-day retention)
2 — Cross-branch	Partner branch NAS via SD-WAN	rsync with encryption	Daily sync	NAS snapshots
3 — Cloud	Backblaze B2	Restic with AES-256	Weekly full + daily incremental	Object Lock (30-day)

10. Remote Access & Management

Secure Remote Access (WireGuard VPN)

Remote users would connect via WireGuard VPN running on each FlexiWAN device. WireGuard is built into all modern operating systems and uses key-based authentication — there are no passwords to brute-force.

Feature	Detail
Protocol	WireGuard (modern, minimal, kernel-level)
Authentication	Public/private key pairs — no passwords
Encryption	ChaCha20-Poly1305 (or AES-256-GCM with HW acceleration)
Split Tunneling	Only branch traffic through VPN; internet browsing stays local
VPN Subnet	10.10.10.0/24 (dedicated, isolated from all branch VLANs)
Client Software	Built into Windows, macOS, iOS, Android, Linux

Remote Administration

The administrator can manage the entire infrastructure remotely through five web dashboards:

Dashboard	What It Manages	Access
flexiManage	All SD-WAN devices, tunnels, firewall rules	Web browser, anywhere
Cloudflare Gateway	DNS policies, block/allow rules, query logs	Web browser, anywhere
ESET PROTECT	Antivirus status across all servers	Web browser, anywhere
Uptime Kuma	Monitoring dashboard, alert configuration	Web browser via VPN
WireGuard VPN	Direct access to any branch for RDP, troubleshooting	VPN client

No on-site presence needed for day-to-day management. All five interfaces are accessible from any location with an internet connection.

11. Ransomware Risk Assessment

This matrix maps every significant attack vector against the current exposure and proposed mitigations.

#	Attack Vector	Current	Proposed Mitigation	After
1	RDP Brute Force	CRITICAL	FlexiWAN blocks all inbound. RDP via VPN only.	ELIMINATED
2	SQL Exploitation	CRITICAL	FlexiWAN blocks inbound. SQL bound to VLAN 10.	ELIMINATED
3	Phishing / Email	HIGH	Cloudflare Gateway + SPF/DKIM/DMARC + training.	MEDIUM
4	Malicious Downloads	HIGH	Cloudflare DNS + Defender + ESET + AppLocker.	LOW
5	Lateral Movement	CRITICAL	VLAN segmentation + Windows Firewall + separate accounts.	LOW
6	Credential Theft	HIGH	VPN uses keys (no passwords). Account lockout. Separate admins.	LOW
7	Backup Destruction	CRITICAL	3-2-1 backup + NAS snapshots + B2 Object Lock.	LOW
8	DNS Attacks	MEDIUM	Cloudflare: DNSSEC + malware blocklist + DGA detection.	LOW
9	Unpatched Vulns	HIGH	Automated Windows Update. Monthly review.	LOW

12. Implementation & Emergency Reference

Proposed Timeline

Week	Scope	Deliverable
1	Hardware procurement + Ubuntu + FlexiWAN installation	4 flexiEdge devices registered in flexiManage
2	Branch 1 pilot — VLANs, firewall, Cloudflare DNS, WireGuard	Master branch fully segmented and protected
3	Deploy branches 2–4 + auto-create full-mesh tunnels	All branches connected. SQL sync encrypted.
4	Server hardening + AV + backup setup	All servers hardened. Daily backups running.
5	Monitoring + offsite backup (Backblaze B2)	Uptime Kuma live. 3-2-1 backup complete.
6	Testing, verification, documentation, training	All tests passed. Runbooks delivered.

Estimated deployment: 6 weeks. Each phase is independently valuable — the network becomes progressively more secure from week one.

Emergency Quick-Reference

Suspected ransomware: ① Isolate — pull Ethernet immediately ② Do not reboot (preserves forensic evidence) ③ Check other branches via flexiManage dashboard ④ Restore from most recent clean backup ⑤ Investigate via Windows logs ⑥ Harden the exploited vulnerability

FlexiWAN device failure: Branch loses connectivity. Boot standby N100 — already registered in flexiManage, auto-configures on startup (~5 min recovery).

SD-WAN tunnel down: Check flexiManage dashboard (accessible from anywhere). Wait 2 minutes for auto-recovery. If still down, check remote branch broadband. Restart tunnel from dashboard — no SSH needed.